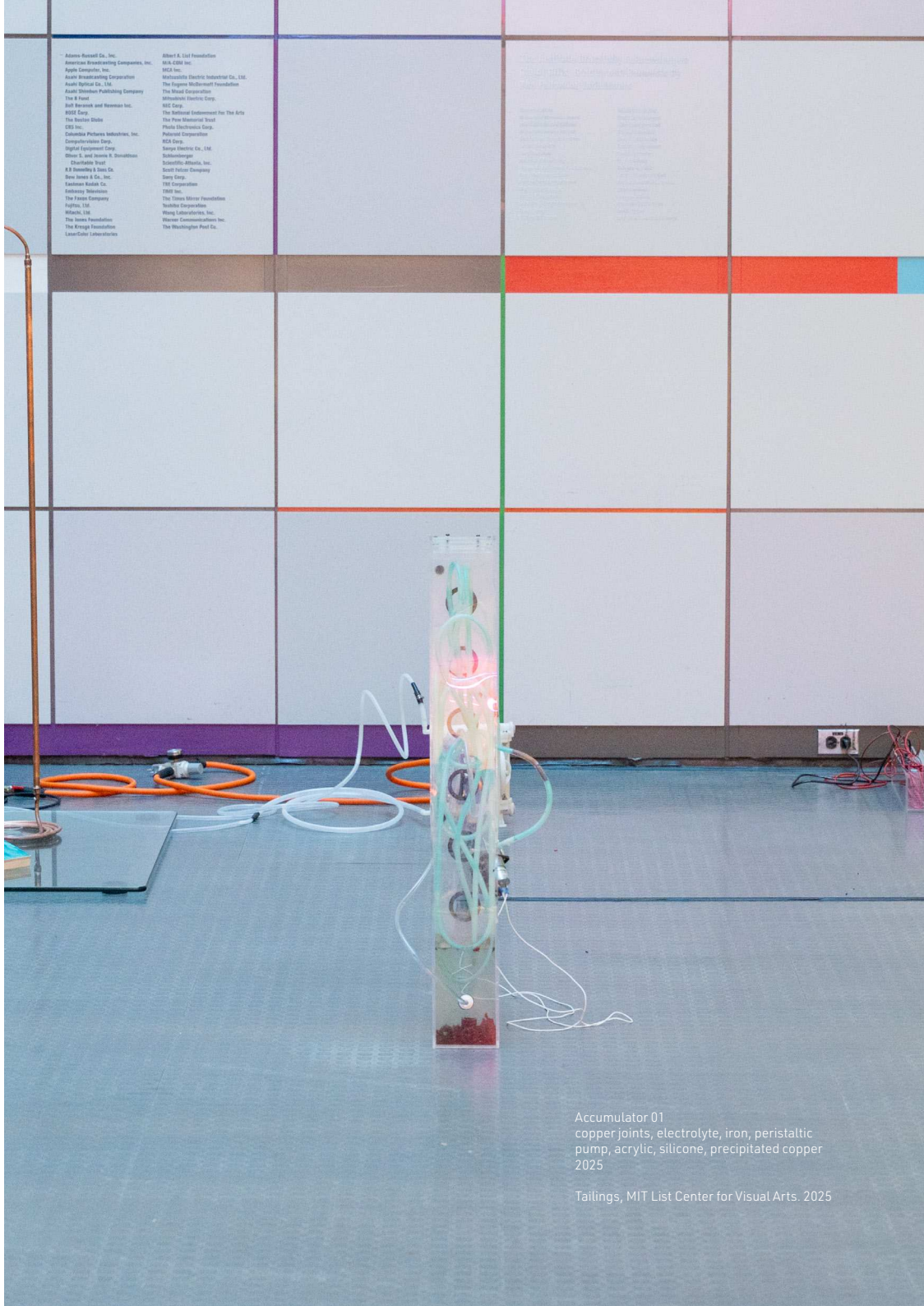
Tailings: New and Old Metabolics in Butte, Montana (Draft)

Accumulated mine tailings from centuries of copper, cobalt, and coal extraction are increasingly recognized by the United States government as latent deposits of rare earth elements—resources now critical to data centers, satellite infrastructure, and military technology.[1] The acidic runoff from these mines is strong enough to dissolve metals from surrounding soil at very high concentrations. While many of these mines have shuttered under global economic pressures, and their laborers either automated or abandoned, these environmental conditions continue to devastate surrounding life. Today, these environmental catastrophes, long characterized as waste, are being recharacterized as valuable domestic orebodies, and sites at the frontier of American technical infrastructure.

These stores of industrial pollution are the subject of a widespread mapping and profiling effort by the US Geological Survey's (USGS) Earth Mapping Resources Initiative (Earth MRI) in its search for "unconventional sources of valuable minerals."[2] The initiative emerges from US anxiety over supply chains strained by fracturing global capitalism and rising conflict, which threaten the minerals AI data centers depend on.[3] With only one domestic source of rare earth elements, the Department of Defense is funding demonstration projects across the nation's interior that experiment with new ways to refine and harvest these second-order residues, from chemical leaching to bioaccumulation and nanoscale filtration.[4]

Tailings grounds itself in the material processes and metabolic flows of Butte, Montana—a storied labor town, and one such demonstration site identified by the USGS and Department of Defense. Once the epicenter of global copper production, Butte is now home to the Berkeley Pit: a polluted, highly acidic lake, and popular example of environmental catastrophe at the hands of industrial capitalism.[5] Recent US efforts to secure a domestic supply of what it terms 'critical minerals' have identified Butte's waterbodies as an unexpected national asset. As virtual space accelerates from massive speculative investments in technical infrastructure, nodes of this buildout surface with unusual clarity in Butte: century-old mine tailings are re-mined for rare earth elements, a 400-MW data center is built along Silver Bow Creek, and Butte's long history of organized labor attempts to orient itself to a new wave of corporate power, environmental recklessness, and automation.[6]

1. U.S. Department of Energy, Critical Minerals and Materials Strategy (Washington, DC: DOE, January 2021).
2. U.S. Geological Survey, "Earth MRI Progress," Earth Mapping Resources Initiative, https://www.usgs.gov/special-topics/earth-mri/science/earth-mri-progress.
3. CHIPS and Science Act of 2022, Pub. L. No. 117-167, 136 Stat. 1366 (2022); U.S. Department of Energy, Critical Minerals and Materials Strategy (2021).
4. https://www.war.gov/News/Releases/Release/Article/3890176/department-of-defense-awards-656-million-to-expand-workforce-development-in-ext/

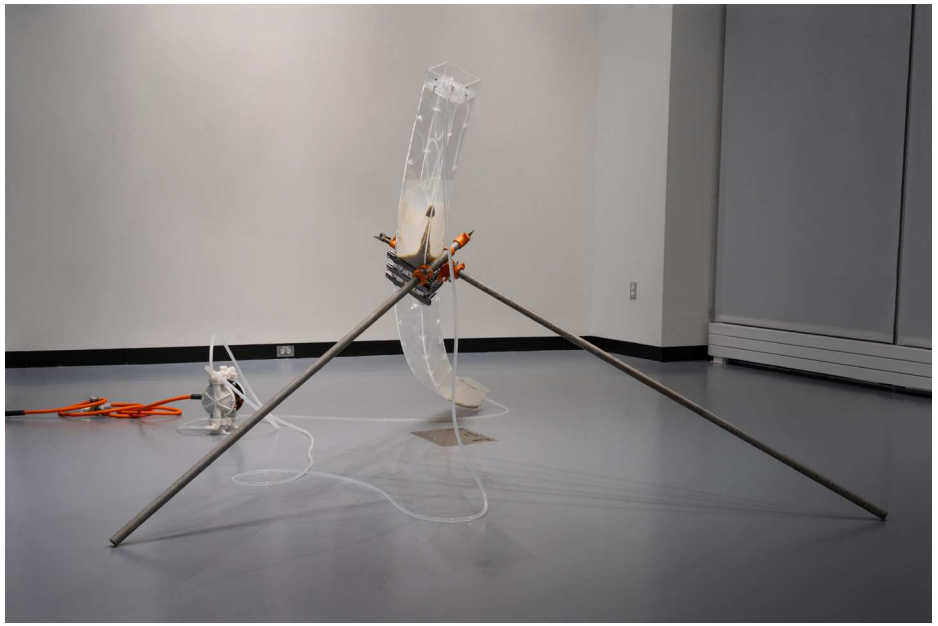


Accumulator 01
copper joints, electrolyte, iron, peristaltic pump, acrylic, silicone, precipitated copper
2025

Tailings, MIT List Center for Visual Arts. 2025

As the models that produce our digital infrastructure begin to consume this second-order material, the model of extraction folds inward, further entering its own synthetic space. Through staged circulatory sculptures and nanoscale imaging of polluted wastewater, *Tailings* explores the logics of industrial capitalism, the mechanical gaze turning inward, loops, false loops, and the component parts of a metabolic model churning.

The project centers a series of arched tanks. These *Accumulators* operate as various "metabolic quotations" of the Butte landscape and citations of the economic models directing its governance and use. The project takes self-referentiality—a folding inward—as both a guiding prompt for artistic research and aesthetic cue.



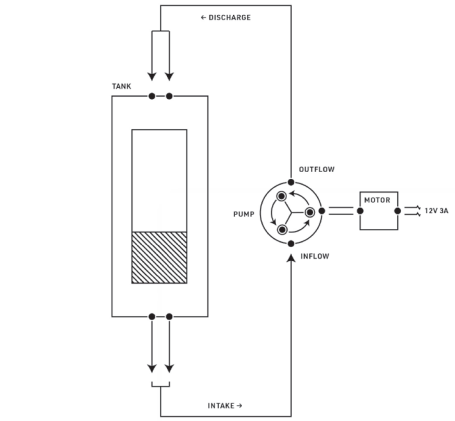
Accumulator 03
acrylic, steel, silicone, diaphragm pump,
bentonite drilling clay, air compressor
2026

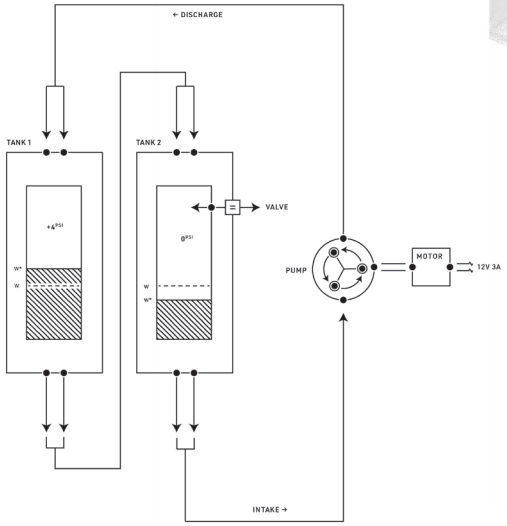
5. U.S. Environmental Protection Agency, "Berkeley Pit," Silver Bow Creek/Butte Area Superfund Site, available at epa.gov/superfund/berkeley-pit; Christopher H. Gammons and Terence E. Duaime, "The Berkeley Pit and Surrounding Mine Waters of Butte," in Geology of Montana, vol. 2: Special Topics, Montana Bureau of Mines and Geology Special Publication 122 (2020).

6. "OpenAI's Nvidia, AMD Deals Boost $1 Trillion AI Boom With Circular Deals," Bloomberg, October 7, 2025, https://www.bloomberg.com/news/features/2025-10-07/openai-s-nvidia-amd-deals-boost-1-trillion-ai-boom-with-circular-deals.



Accumulator 03
acrylic, steel, silicone, diaphragm pump,
bentonite drilling clay, air compressor
2026



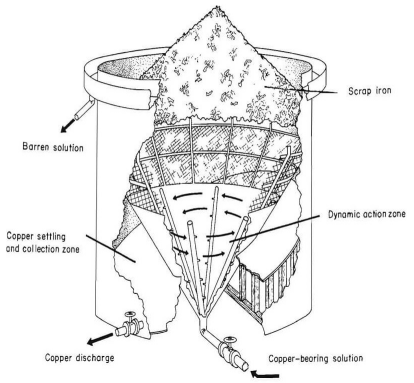




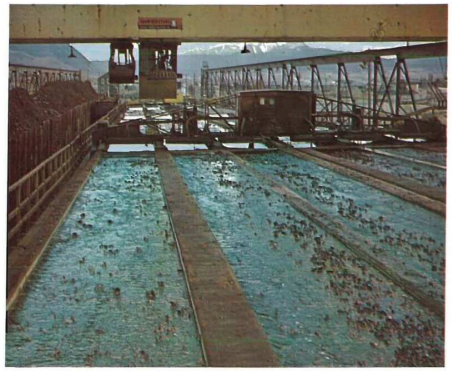
Models and flow diagrams of the Accumulator series' programmable tanks. One tank is a scalable base unit used for this series of sculptures.

* system at steady state

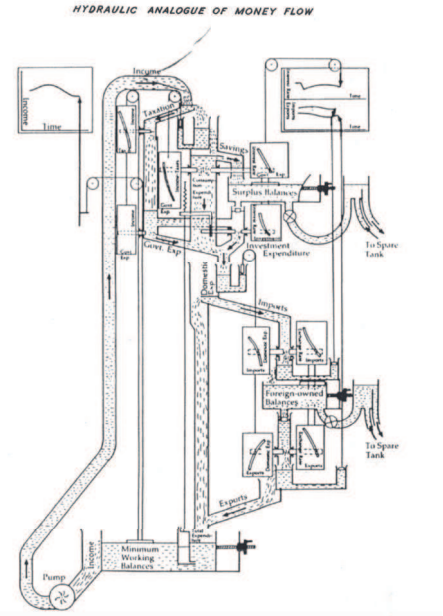
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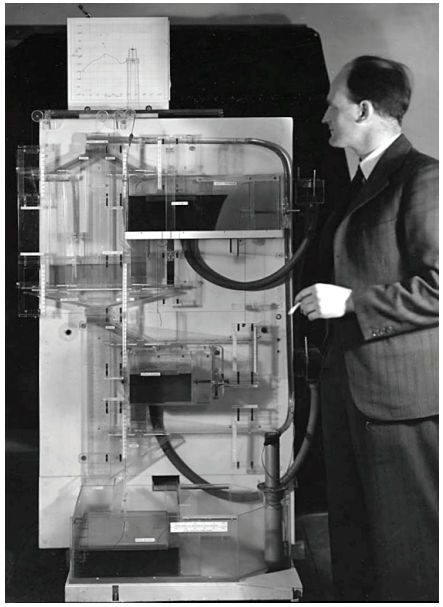
[Figure 1] Cutaway Diagram of Cone Precipitator for the reclamation of copper from Kennecott Copper Corp.



[Figure 2] The Horseshoe Bend Water Treatment Facility in Butte, Montana is operated by the Montana Resources. The facility pumps copper-rich lake water from the Berkeley Pit onto large baths containing bits of scrap iron and steel. Through the process of cementation, the dissolved copper is released as solid nodules that are then collected for smelting and sale.



[Figure 3] The Phillips Economic Analog Computer, dubbed the MONIAC: Monetary National Income Analogue Computer, as a reference to ENIAC, early digital computer that ran the first computerized weather forecast a year earlier.



[Figure 4] Inventor and economist Bill Phillips, demonstrates in a physical way the circular flow of money within the economy.

Accumulator 01 and *Accumulator 04* cite a historic metallurgical process for recovering copper from acid mine drainage. (Figures 1, 2) The method relies on a chemical reaction between dissolved copper and iron, used by mining companies and environmental remediation groups since the early 20th century. When copper-laden mine water contacts iron, the dissolved copper cements, or plates, onto the iron in metallic form. This redox reaction grows copper nodules that are economical for collection, smelting, and sale. This technique is still in operation in the Berkeley Pit mining complex today.

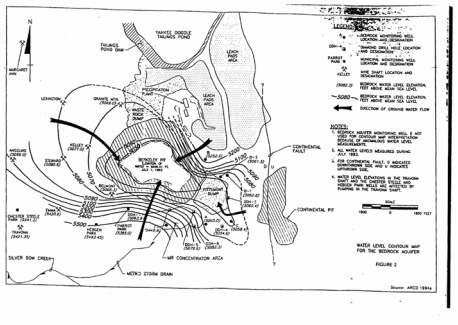
The project also references a lineage of hydrologic machines built to compute and forecast economic flows. In particular, the works in *Tailings* reference the Phillips Machine (Figure 3,4), also known as the MONIAC, a name that nods to the ENIAC, the early digital computer that ran the first computerized weather forecast a year earlier. The machine was conceived by Bill Phillips, an engineer turned economist, to demonstrate the circular flow of capital in a national economy.[7] Income, taxation, savings rates, and consumption curves were each encoded in a physical mechanism: a valve, pump, tank, or channel. Its command of environmental media: water, but also air and electricity, exemplifies the modern fantasy of Keynesian levers, and with it the technical fantasy of orchestrating the flow of environmental media across landscape.

Contemporary proposals to re-mine Butte's industrial residues for rare earth elements continues a long legacy of economic, imperial, and technical forces driving the extraction of this ground in deeper and more expansive ways.

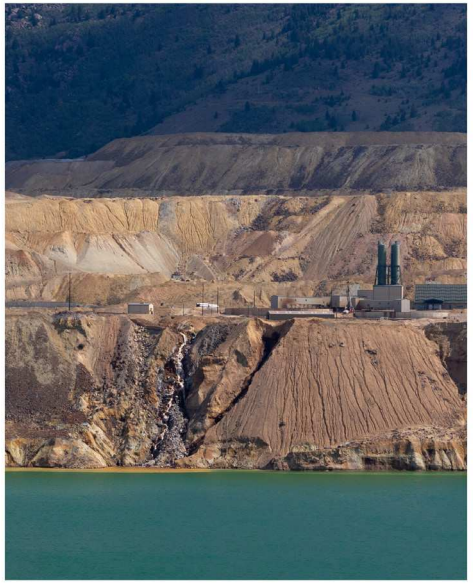
Mining began in Butte in the late 19th century, and the demand for copper to electrify the United States grew its consolidated mining corporations into the nation's largest copper producers within a matter of decades. The Anaconda Copper Mining Company would eventually monopolize the site, pumping and ventilating an expansive network of shafts and adits to extract ore from beneath the city. By the 1950s, as ore grades declined and global competition rose, the company turned to a different model of efficiency. In 1955 it opened the Berkeley Pit, an open-pit operation that profits from economies of scale: moving vast volumes of low-grade ore and concentrating it through chemical and mechanical means. Under this logic the ground beneath the city became more valuable than the city itself, and the pit expanded to consume entire neighborhoods, including Meaderville, East Butte, and McQueen.[8]

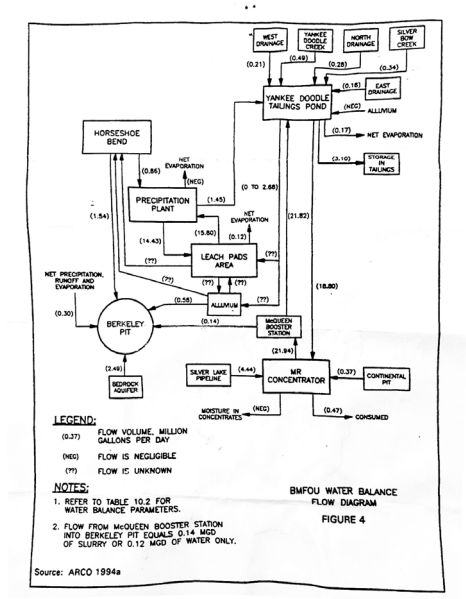
In 1977 the Atlantic Richfield Company (ARCO) acquired Anaconda, folding the mine into the portfolio of an oil conglomerate. Five years later ARCO suspended operations and shut off the pumps. Soon groundwater began to rise through the abandoned workings and into the pit. These rising water levels prompted a different form of management: a managed technical landscape [9], held in equilibrium in perpetuity, an apparatus of containment through which the state preserved itself against liability.





[Figure 5] In 1994 the EPA fixed a Protective Water Level of 5,410 feet, fifty feet below the elevation at which contaminated water would begin to flow out of the pit and into the surrounding aquifer and Silver Bow Creek. Today, the Berkeley Pit must remain the lowest point in Butte's hydrology, a permanent sink maintained by a vast complex of pumps, siphons, and ponds. Its water is now pumped, stripped of copper, neutralized, and released into Silver Bow Creek.





[Figure 6] U.S. Environmental Protection Agency, "Amendment to National Oil and Hazardous Substance Contingency Plan; National Priorities List," Federal Register 48, no. 175 (September 8, 1983): 40658–73.

Today's proposals to re-mine the mining residues of Butte are advanced by a warped version of these economic models and imperial state power. The extreme rationality of Phillips' MONIAC looks even more of a pipe-dream than ever before. No, what really seems to be a force in the governance of ground today is a different order of capital: speculative, self-referential, and increasingly abstract. A high frequency swarm of speculative positions taken on positions, bound and structured by state and corporate power.[10]

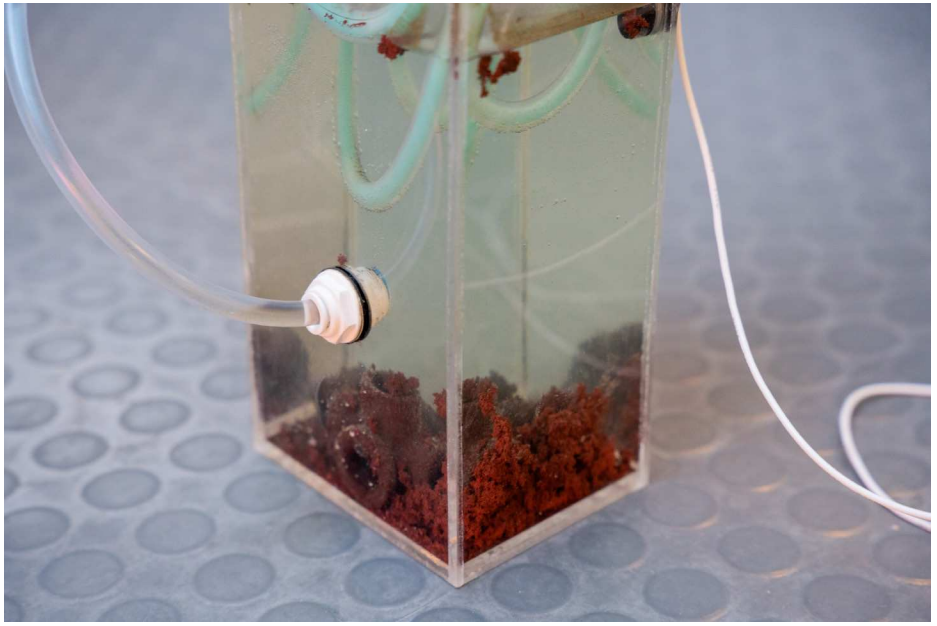
The forces directing investment in Butte look much more abstract and more volatile. Butte, and sites like it across the nation's interior, are the material endpoint supplying state and corporate speculation on the shape and material resource demands of a new economy emerging from old: a bet on the mass automation of software and information economy. A bet on capital over human labor. These investments measure in the hundreds of billions of dollars, and flow through a new set of corporate monopolies. Each holding a share in the other, each modeling a software economy that is increasingly self-referential. To the degree these trends are bounded by the state, it is as investor rather than regulator.

If the MONIAC imagined the economy as water moving through tanks, this newer economy is a version of that machine warped beyond its own legibility. In *Tailings*, the tank extends a version of this mirrored hydrologic metaphor: a fantastical dual image for the governance of economy and environmental flow. Referents collapse into loops and sediments that oscillate between model and thing. These arched forms cycle together as an orchestra.

7. A.W.H. Phillips, "Mechanical Models in Economic Dynamics," Economica 17, no. 67 (1950): 283--305.
8. The City That Ate Itself: Butte, Montana and Its Expanding Berkeley Pit (Reno: University of Nevada Press, 2018).
9. U.S. Environmental Protection Agency, "Amendment to National Oil and Hazardous Substance Contingency Plan; National Priorities List," Federal Register 48, no. 175 (September 8, 1983): 40658–73.
10. Esposito, The Future of Futures.

Accumulator 01
copper joints, electrolyte, iron, peristaltic pump, acrylic, silicone, precipitated copper
2025

Tailings, MIT List Center for Visual Arts. 2025





Top: Lower bath of *Accumulator 01* with cemented copper produced from the tank's circuitry.

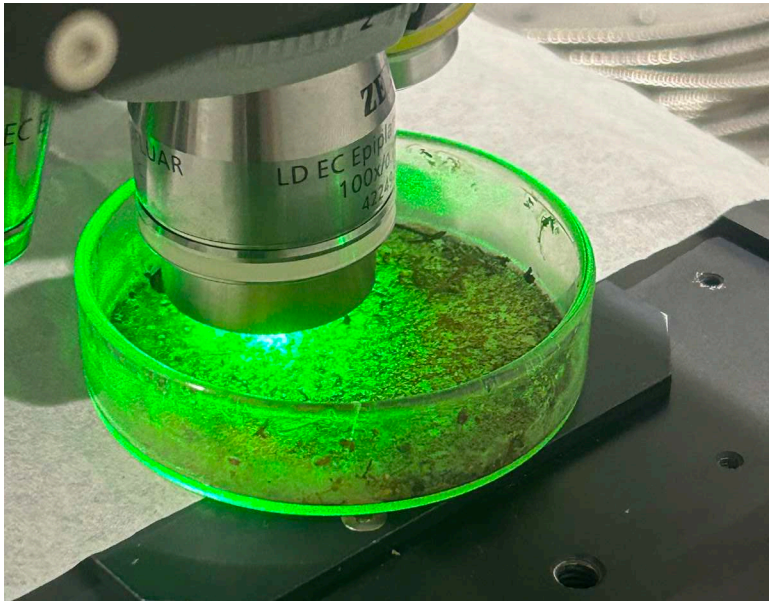
Bottom: Lower bath of *Accumulator 01* with cemented copper after several weeks.



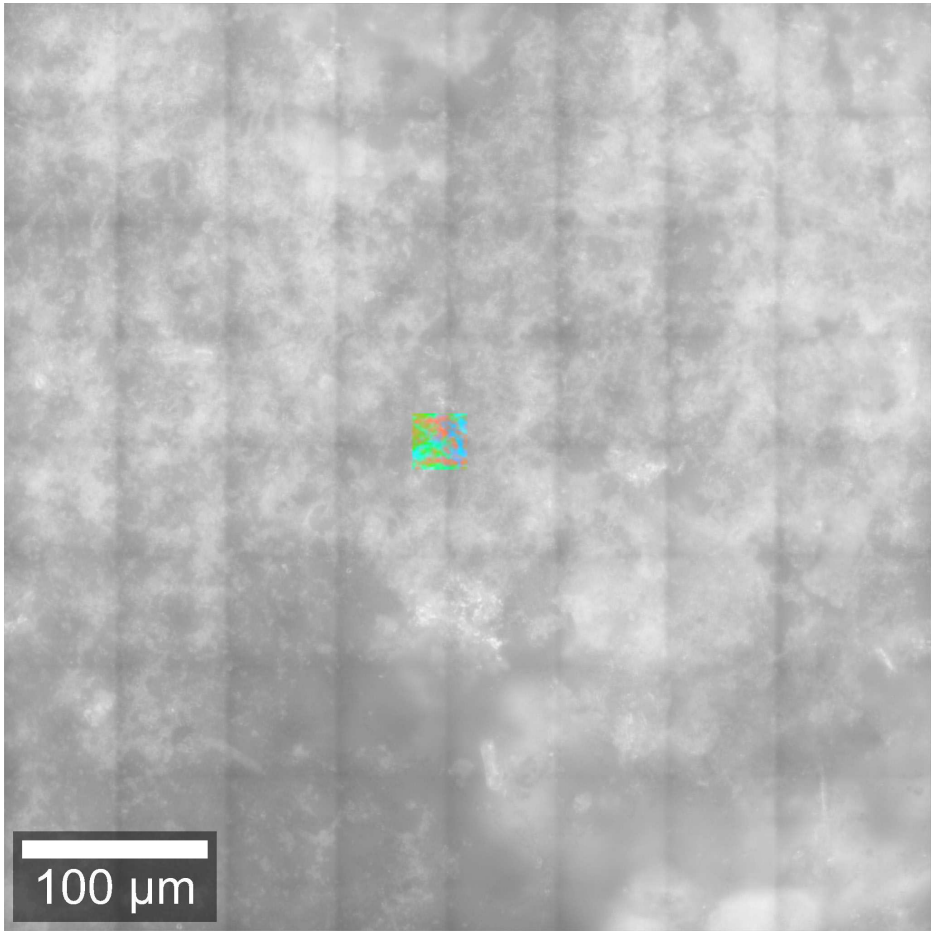
Live embedding representations of an autoencoder machine learning model training on archival image dataset of Butte, Montana.



[Figure 5] Petridish sample of cement copper from of *Accumulator 01*.



[Figure 6] RAMAN spectrograph of wastewater sample from Superfund Site in Woburn MA taken by the artist. This method of imaging maps the various chemical compositions of sample material. RAMAN spectroscopy is used in a variety of fields, from the characterization of paint in art conservation, to the search heavy metals in environmental samples.



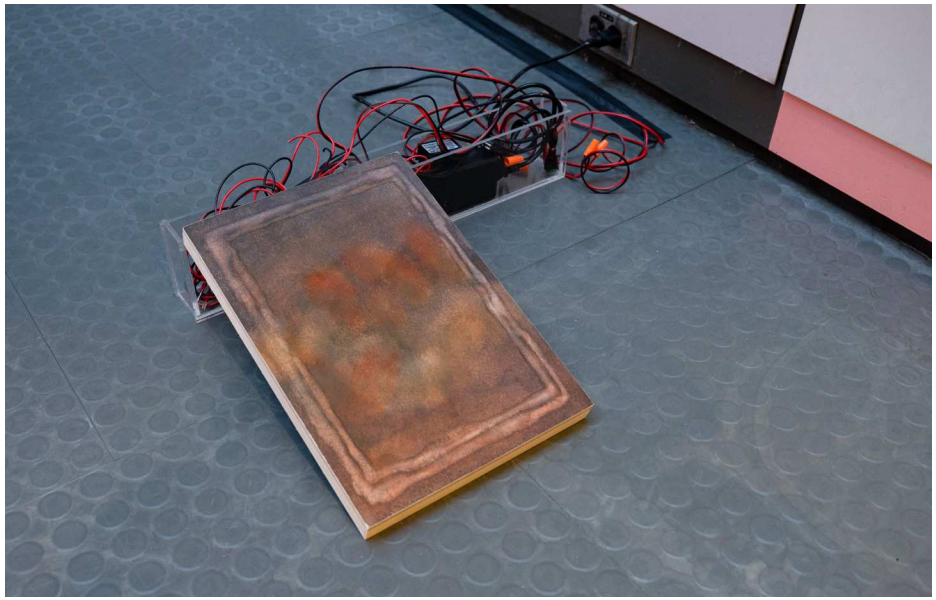
[Figure 6] Gradient map of materials in industrial wastewater sample. Composite image was captured and configured by confocal Raman imaging system alpha300 from Witec.

This body of artistic research (networked tanks, generative video, photography, and archival work) frames a landscape produced by economic, political, and social forces over time. Its contours record an accumulation of sediment formed in the gaps between model and ground. In Butte, that ground is shifting again, reoriented toward industrial production.

In some type of personal refusal, I collect the copper sediments that form in *Accumulator 01* and *Accumulator 04*, grind them into pigment, and bind them into watercolor with honey and gum arabic. These rough paints join verdigris and other pigments made in the studio, and are used in a series of paintings (see Untitled and Model 15 – Berkeley Pit).

The paintings become a kind of metabolic image.

I develop these works by iteratively lifting and washing paint in relation with the emerging network of marks. The works digest experiences from walks and specific reference imagery. This process surfaces larger forms that evoke swirling ecologies and other networked things. Within this cycle, interventions are introduced, reconciled, and consumed until the composition reaches visual equilibrium.



Untitled
11 x 17 inches
copper and cobalt derived pigments on watercolor on paper, mounted on wood panel
2025



Model 14
41 x 30 inches
watercolor on paper
2025



Model 15 (berkeley pit)
30 x 7.5 inches
copper and cobalt derived pigments on watercol-
or on paper, mounted on wood panel
2026



Model 9
21 x 6 inches
watercolor on paper, mounted on wood panel
2025



Plots, Grids, Models
Kirkland Gallery, Harvard University
2026

Eric Rannestad explores how environmental models, economy, and visual culture co-produce landscape. The work draws on the language of maps, forecasting models, capital flows, technical instruments, and other site-specific artifacts to investigate themes of uncertainty, ground, and circulation. Working through both designed process and metabolic gesture, his projects combine media that span excavated watercolor surfaces, arched sculptural components, time-based, and generative media: each a different cut in the interrelated forces that construct landscape, each offering a different register for locating oneself within these forces.

Eric Rannestad (b. 1996 Chester, CT) lives and works in Cambridge, MA. He received a Master of Design Studies (MDes) from the Harvard Graduate School of Design and a BA in Studio Art and Economics from Whitman College. His work has been supported by the Massachusetts Cultural Council, with fellowships from the Vermont Studio Center and I-Park Foundation and residencies at the Sitka Center for Art and Ecology and the Wassaic Project, among others. He has exhibited at Chilli Gallery in London, the Emerson Center for the Arts & Culture in Bozeman, Montana, and NADA New York, and his work has been featured in New American Paintings. Eric's projects in cartography and experience working with environmental groups across the American West are a significant influence in his art practice.